

Confidence Intervals, Regions and Prediction Bands

A test asks whether one value of a parameter is compatible with the data; an interval reports all such values, and how precisely they are pinned down. Under normal errors the answers are exact for every sample size, and they come from three facts: the fitted vector is normal, the residual sum of squares is an independent chi-squared variable, and their ratio gives t and F pivots. One estimable function gives a t interval; several give an ellipsoid, the dual of the F test. A future observation gives a prediction interval, whose width never falls below the noise and whose validity rests on the normality of a single error. The whole regression surface gives the Working–Hotelling band, from the Cauchy–Schwarz inequality. Inverting a prediction band gives a calibration set, sometimes an interval, sometimes two rays, sometimes the whole line, and no honest procedure can avoid the last two. Finally, in the conjugate model of [Chapter 7](#), credible and predictive intervals have the same shapes, and under the flat prior they are the frequentist ones.

What you need. The sampling distributions of the normal linear model ([Chapter 7](#): Theorem 7.5.1, Corollary 7.5.2 and Corollary 7.5.3) and the conjugate posterior (Theorem 7.6.2, Lemma 7.6.1, Corollary 7.6.4). Projections, leverage and the Frisch–Waugh–Lovell theorem ([Chapter 6](#): Theorem 6.3.6, Proposition 6.8.1, Theorem 6.6.2). Estimability ([Chapter 8](#): Definition 8.2.1, Theorem 8.3.1) and the sum of squares of a linear hypothesis ([Chapter 9](#): Theorem 9.2.4). Chi-squared and F distributions of quadratic forms ([Chapter 4](#): Theorem 4.3.2, Definition 4.1.2, Definition 4.1.3). The linear model itself, with the moments of $\hat{\beta}$ and the consistency of s^2 , comes from [Chapter 5](#) (Theorem 5.5.1, Corollary 5.5.4, Eq. 5.5.2, Corollary 5.6.4), and the normal distribution theory the intervals are read off — the density, the behaviour under a linear map and the independence of uncorrelated blocks — from [Chapter 3](#) (Theorem 3.2.2, Theorem 3.3.1, Theorem 3.4.1); [Section 12.2](#) also uses the generalized Rayleigh quotient of [Chapter 1](#) (Corollary 1.7.8). The tests of [Chapter 11](#) are the duals of the intervals here, but the proofs below do not depend on them.

12.-1.1 Roadmap

- [Intervals for estimable functions](#)
- [Confidence ellipsoids](#)
- [Prediction intervals](#)
- [Simultaneous bands for the regression line and surface](#)
- [Inverse prediction and calibration](#)
- [Bayesian thread II: credible intervals and predictive distributions](#)
- [Summary and notes](#)